

EXPLORATORY ANALYSIS OF RAIN FALL DATA IN INDIA FOR AGRICULTURE

DATE	28-02-2026
TEAM ID	LTVIP2026TMIDS89121
PROJECT NAME	Exploratory Analysis of Rain Fall Data in India for Agriculture
MAXIMUM MARKS	2MARKS

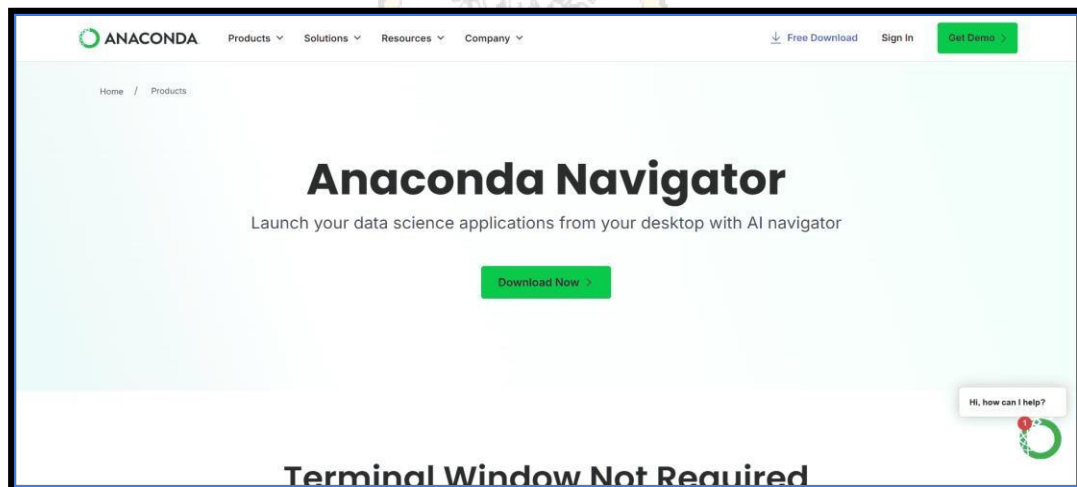
CHAPTER-6 PROJECT DEVELOPMENT

6.1-PRE-REQUISITES:

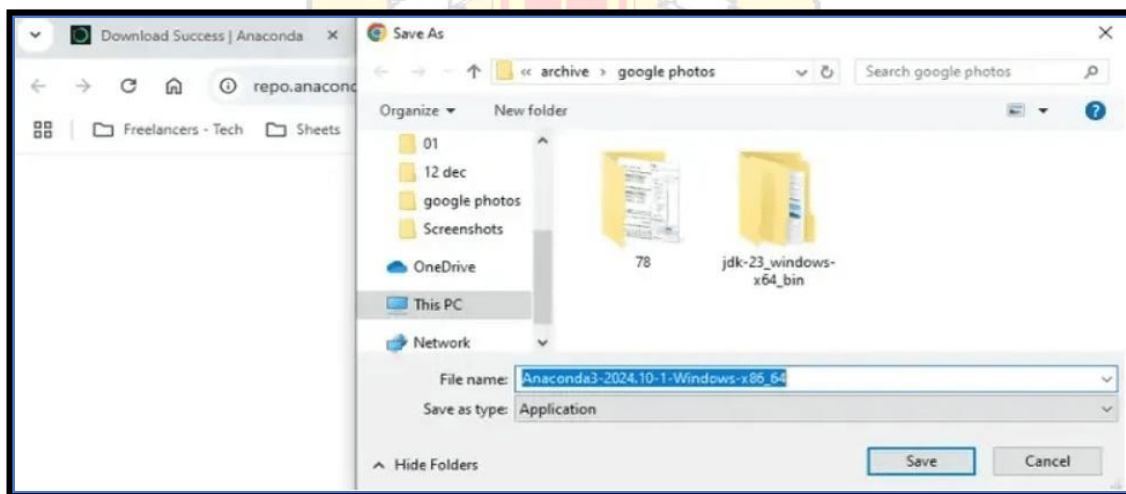
In order to develop this project, we need to install the following software/packages:

Step 1: Anaconda Navigator

- Open this link <https://www.anaconda.com/products/navigator>

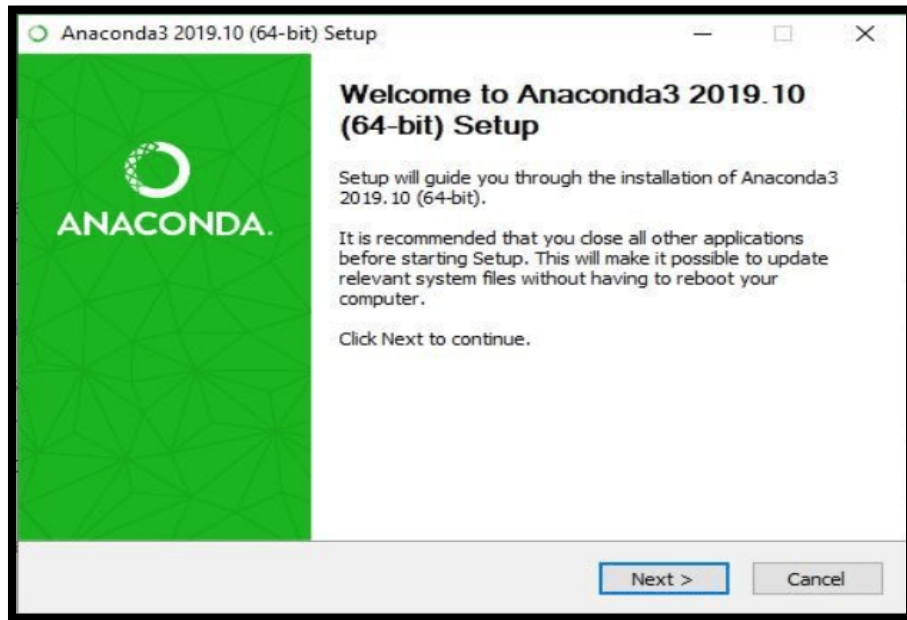


- Click on Download button

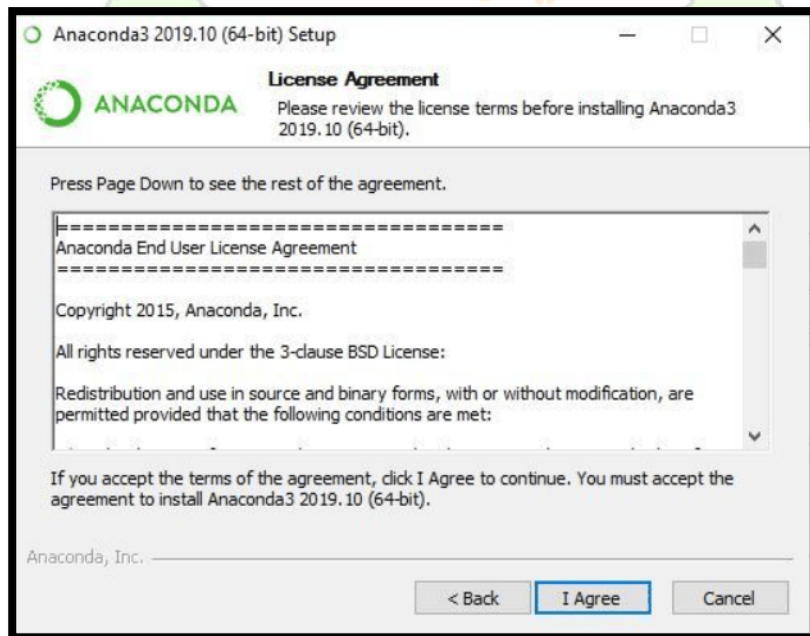


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- After downloading the Anaconda Navigator double click on the file
- Select the Windows Installer
- Begin the installation process.

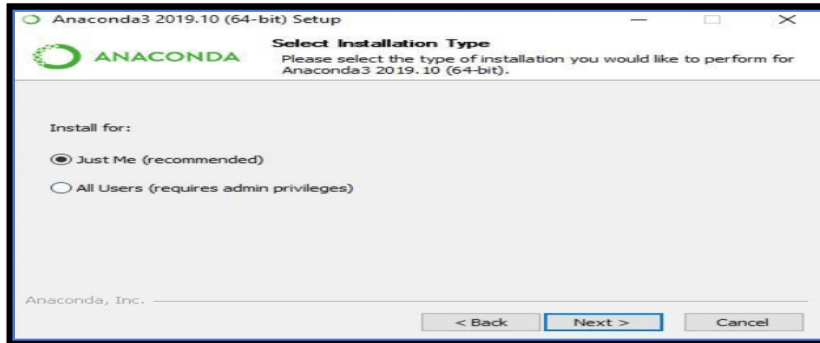


- Getting through license agreement

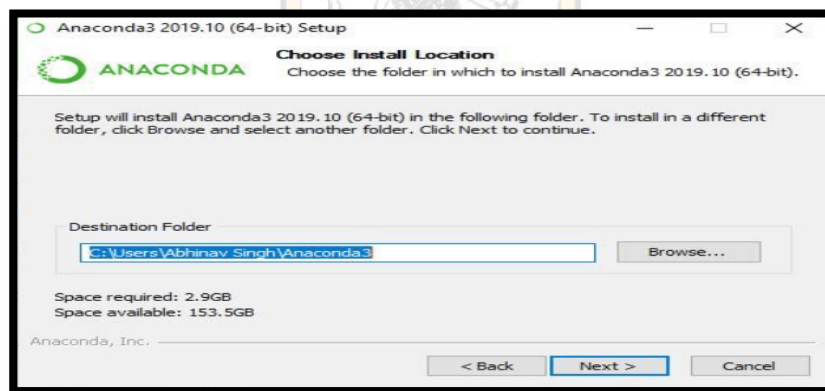


EXPLORATORY ANALYSIS OF RAIN FALL DATA IN INDIA FOR AGRICULTURE

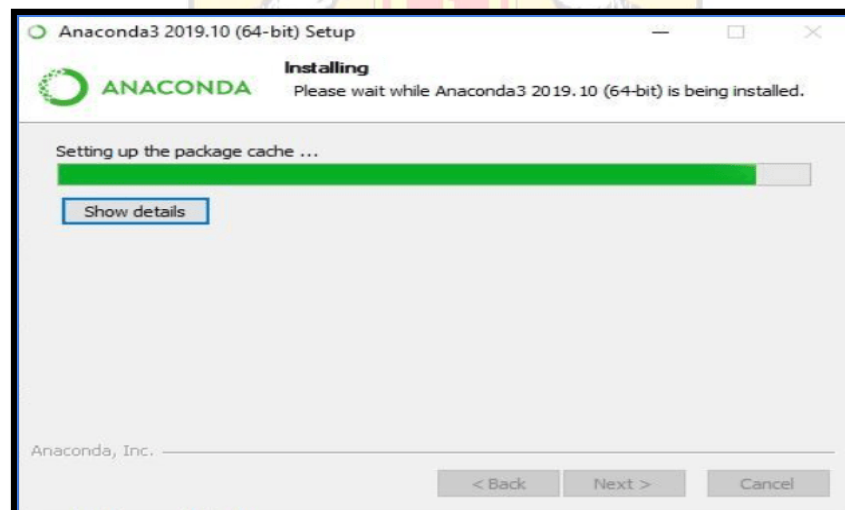
- Select Installation Type



- Select the path where you wish to install the file extract or and click "Next" to proceed ahead.
- Advance installation options

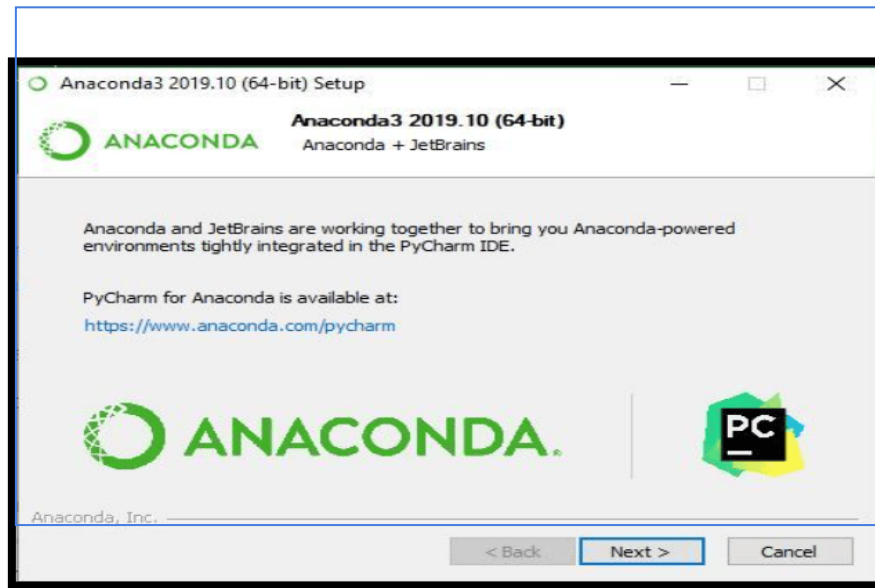


- Click Install to start the Anaconda Installation process.

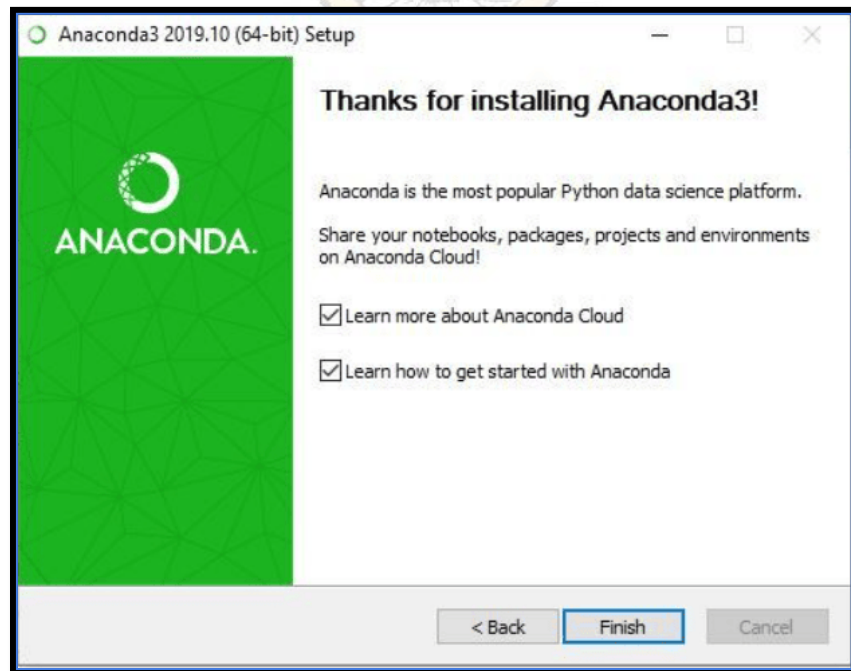


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- Recommendation to install PyCharm

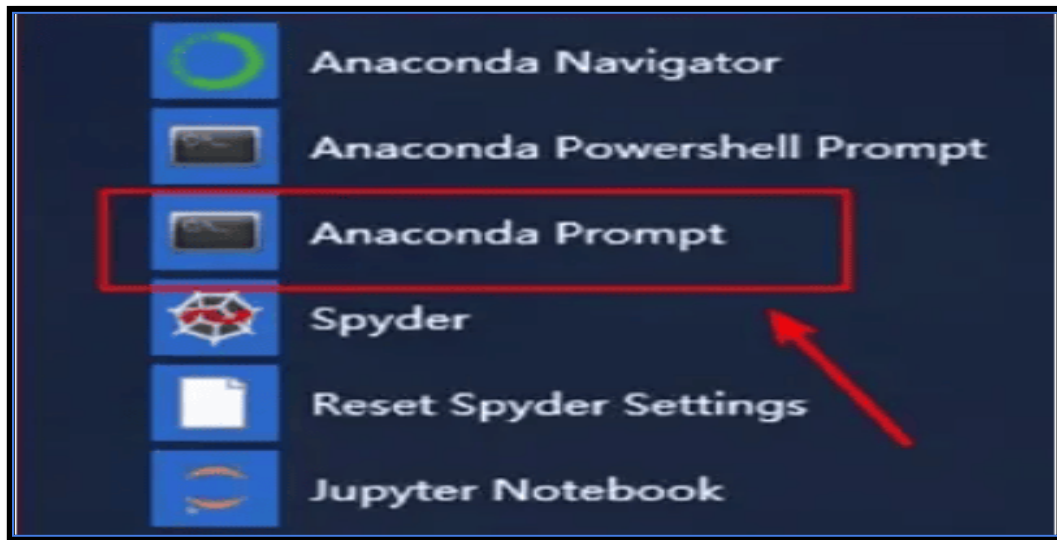


- Once the installation gets complete, Click Finish to complete the process.



- Click on the Start Menu and search for "Anaconda Prompt" and click to open it.

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- Run the Program to check for the Anaconda Version

Type the following command to check for the installed version of conda:

Conda -version

```
Anaconda Prompt

(base) C:\Users\SHREYA>conda --version
conda 23.3.1

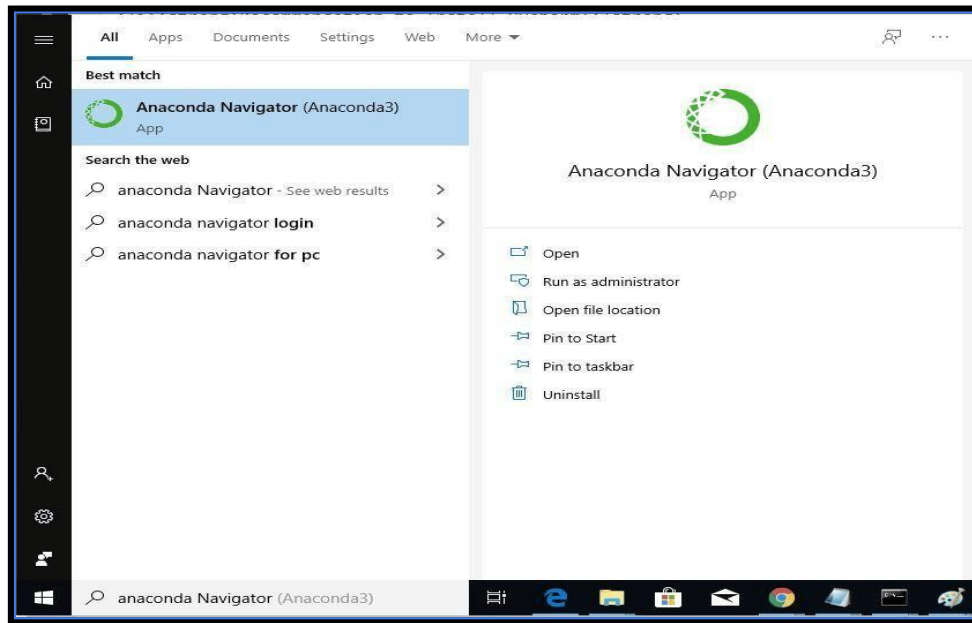
(base) C:\Users\SHREYA>python --version
Python 3.10.9

(base) C:\Users\SHREYA>
```

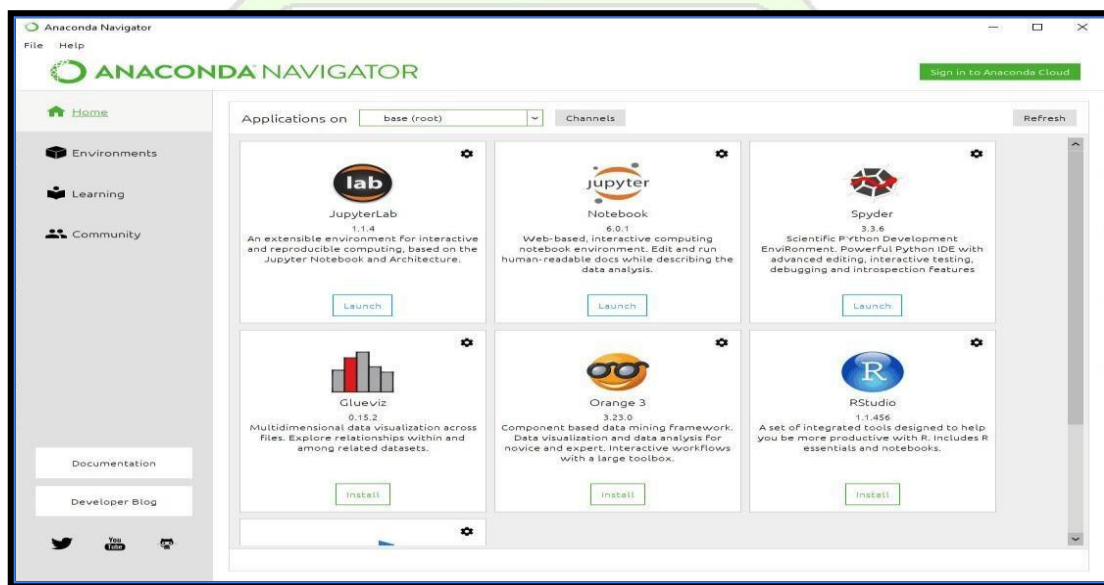
- Access Anaconda Navigator
- Once the installation process is done, Anaconda can be used to perform multiple operations. To begin using Anaconda, search for Anaconda Navigator from the

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Start Menu in Windows PC.



➤ Explore Navigator & Features:



How to Install PyCharm:

Step1: Download PyCharm

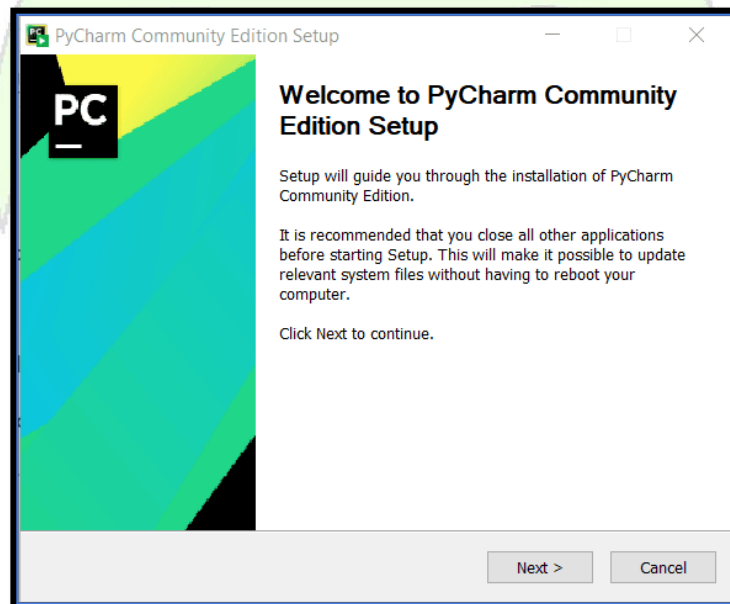
- Go to the Jet Brains PyCharm download page
- Choose either the Community edition(free) or the Professional edition(paid).



- Download the installer according to your preference.

Step - 2: Run the Installer

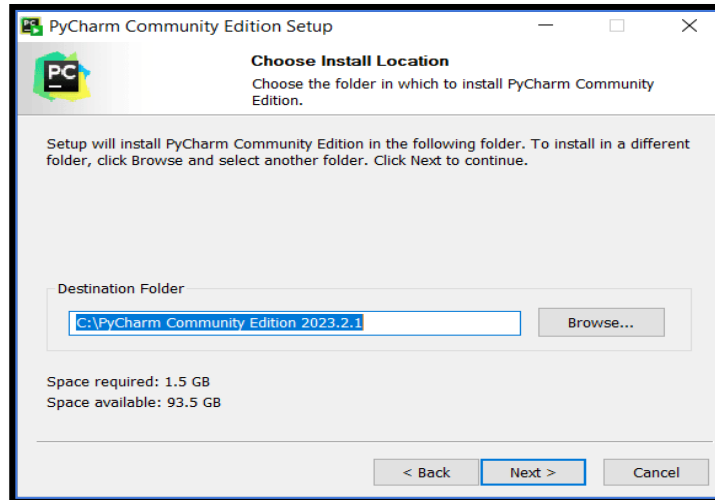
- Locate the downloaded installer file and double-click it to run.
- Follow the on-screen instructions.



Step-3: Choose Installation Location

- Select the folder where you want to install PyCharm.

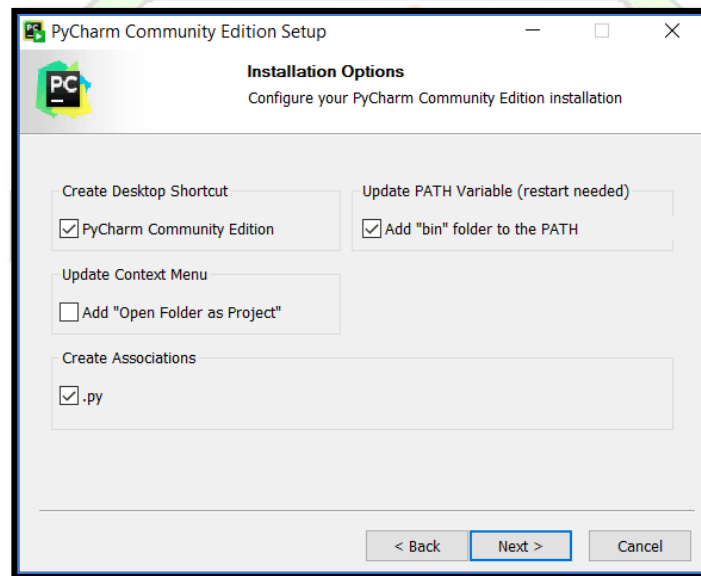
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Step-4: Select Installation Options

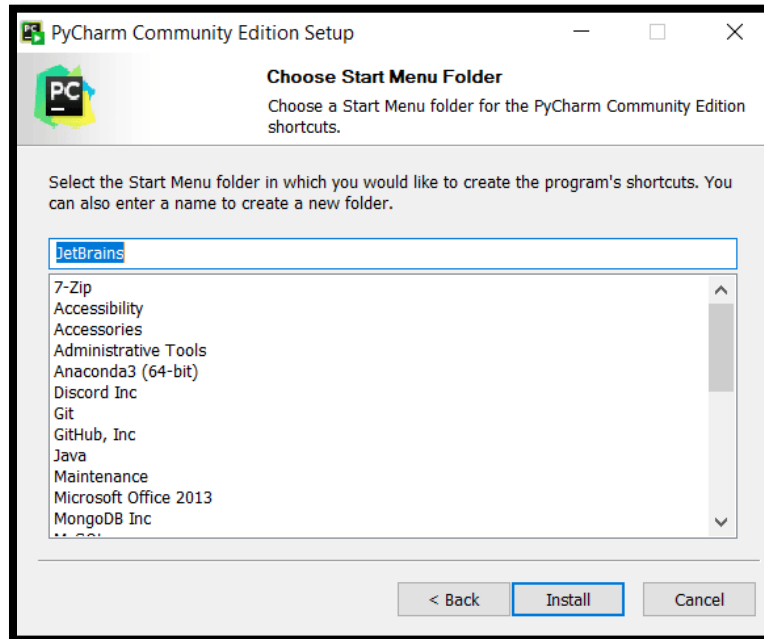
You can select the following options according to your preference:

- Create Desktop Shortcut-adds a shortcut on your desktop.
- Add PyCharm to PATH-allows you to run PyCharm from the command line.
- Associate.py files with PyCharm-makes PyCharm the default editor for Python files.

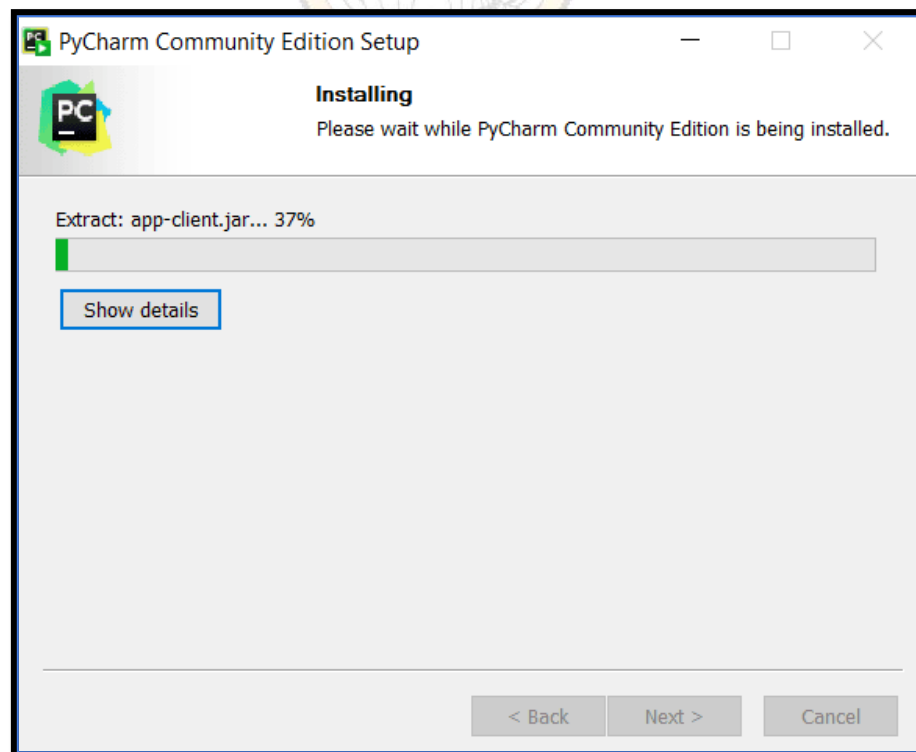


- Choose folder name where your project will be saved as default.

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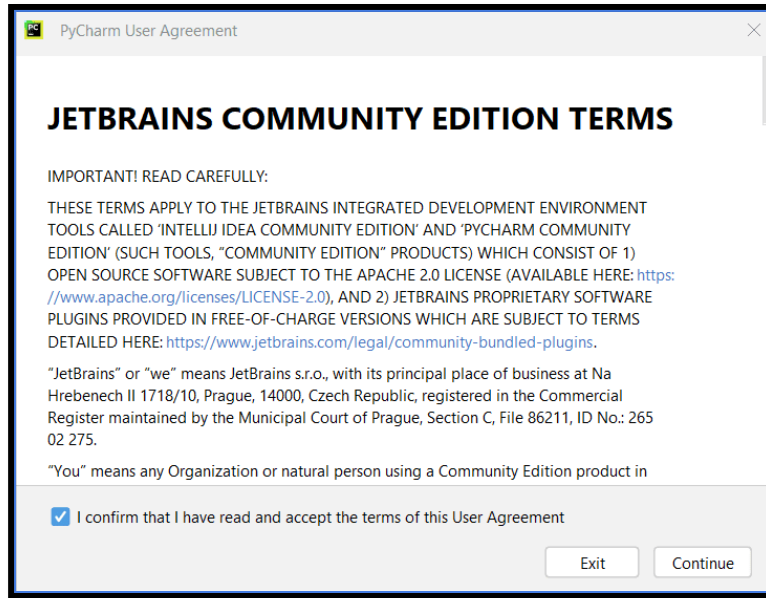
- Now it will download the PyCharm



Step5: Complete Installation

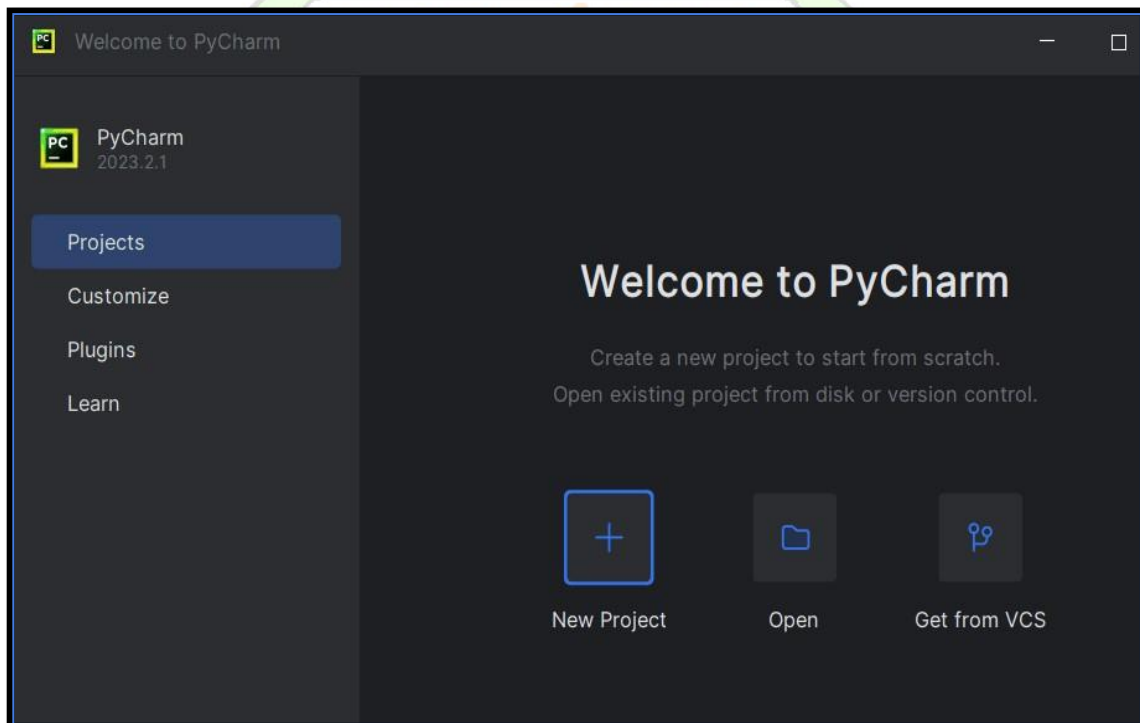
- Accept the license agreement and continue.
- PyCharm will be installed.

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Step6: Launch PyCharm

- Once installed, launch PyCharm either from the Start menu or the desktop shortcut.



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How to install Packages:

Now you are using PyCharm IDE, follow the below steps to download the required packages:

Type “pip install NumPy” and click enter.

```
(.venv) PS D:\LT Python Project> pip install numpy
Requirement already satisfied: numpy in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (2.2.3)

[notice] A new release of pip is available: 25.1.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS D:\LT Python Project> |
```

Like this we have to install all the packages

- Type “pip install pandas” and click enter

```
(.venv) PS D:\LT Python Project> pip install pandas
Requirement already satisfied: pandas in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (2.2.3)
Requirement already satisfied: numpy>=1.26.0 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from pandas) (2.2.3)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from pandas) (2025.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from pandas) (2025.1)
Requirement already satisfied: six>=1.5 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)

[notice] A new release of pip is available: 25.1.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS D:\LT Python Project>
```

- Type “pip install scikit-learn” and click enter.

```
(.venv) PS D:\LT Python Project> pip install scikit-learn
Requirement already satisfied: scikit-learn in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (1.7.2)
Requirement already satisfied: numpy>=1.22.0 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from scikit-learn) (2.2.3)
Requirement already satisfied: scipy>=1.8.0 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from scikit-learn) (1.16.2)
Requirement already satisfied: joblib>=1.2.0 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from scikit-learn) (1.5.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from scikit-learn) (3.6.0)

[notice] A new release of pip is available: 25.1.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS D:\LT Python Project> |
```

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- Type "pip install matplotlib" and click enter.

```
(.venv) PS D:\LT Python Project> pip install matplotlib
Requirement already satisfied: matplotlib in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (3.10.1)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (1.3.1)
Requirement already satisfied: cycler>=0.10 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (4.56.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (1.4.8)
Requirement already satisfied: numpy>=1.23 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (2.2.3)
Requirement already satisfied: packaging>=20.0 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (24.2)
Requirement already satisfied: pillow>=8 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (11.1.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (3.2.1)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\users\hp\appdata\local\programs\python\python313\lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)

[notice] A new release of pip is available: 25.1.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS D:\LT Python Project>
```

- Type "pip install SciPy" and click enter.

```
(.venv) PS C:\LT project> pip install scipy
Requirement already satisfied: scipy in c:\lt project\.venv\lib\site-packages (1.17.0)
Requirement already satisfied: numpy>=2.7, >=1.26.4 in c:\lt project\.venv\lib\site-packages (from scipy) (2.4.2)

[notice] A new release of pip is available: 25.1.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

- Type "pip install pickle-mixin" and click enter.

```
(.venv) PS C:\LT project> pip install pickle-mixin;
Requirement already satisfied: pickle-mixin in c:\lt project\.venv\lib\site-packages (1.0.2)

[notice] A new release of pip is available: 25.1.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

(.venv) PS C:\LT project>
```

- Type "pip install Flask" and click enter.

```
(.venv) PS C:\LT project> pip install Flask;
Requirement already satisfied: Flask in c:\lt project\.venv\lib\site-packages (3.1.2)
Requirement already satisfied: blinker>=1.9.0 in c:\lt project\.venv\lib\site-packages (from Flask) (1.9.0)
Requirement already satisfied: click>=8.1.3 in c:\lt project\.venv\lib\site-packages (from Flask) (8.3.1)
Requirement already satisfied: itsdangerous>=2.2.0 in c:\lt project\.venv\lib\site-packages (from Flask) (2.2.0)
Requirement already satisfied: jinja2>=3.1.2 in c:\lt project\.venv\lib\site-packages (from Flask) (3.1.6)
Requirement already satisfied: markupsafe>=2.1.1 in c:\lt project\.venv\lib\site-packages (from Flask) (3.0.3)
Requirement already satisfied: werkzeug>=3.1.0 in c:\lt project\.venv\lib\site-packages (from Flask) (3.1.5)
Requirement already satisfied: colorama in c:\lt project\.venv\lib\site-packages (from click>=8.1.3->Flask) (0.4.6)

[notice] A new release of pip is available: 25.1.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

(.venv) PS C:\LT project>
```

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- Type” **pip install seaborn**” and click enter.

```
(.venv) PS C:\lt project> pip install seaborn;
Requirement already satisfied: seaborn in c:\lt project\.venv\lib\site-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\lt project\.venv\lib\site-packages (from seaborn) (2.4.2)
Requirement already satisfied: pandas>=1.2 in c:\lt project\.venv\lib\site-packages (from seaborn) (3.0.0)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\lt project\.venv\lib\site-packages (from seaborn) (3.10.0)
Requirement already satisfied: contourpy>=1.0.1 in c:\lt project\.venv\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\lt project\.venv\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\lt project\.venv\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\lt project\.venv\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\lt project\.venv\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (26.0)
Requirement already satisfied: pillow>=8 in c:\lt project\.venv\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (12.1.0)
Requirement already satisfied: pyparsing>=3 in c:\lt project\.venv\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in c:\lt project\.venv\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: tzdata in c:\lt project\.venv\lib\site-packages (from pandas>=1.2->seaborn) (2025.3)
Requirement already satisfied: six>=1.5 in c:\lt project\.venv\lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)

[notice] A new release of pip is available: 25.1.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

